Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics *(L3, L4)*

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

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B. Outlier Detection and Treatment

i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

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C. Categorical Data Standardization

i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*
An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
ii. Interpret whether the relationship is **positive or negative**
iii. Explain how covariance helps in identifying feature relationships

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B. Feature Selection

i. Identify redundant attributes using **correlation or domain knowledge**
ii. Select an optimal subset of features for modeling
iii. Justify your selection

C. Data Transformation

i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics *(L3, L4, L5)*

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

i. Apply **3 bins** for Study_Hours and Test_Score
ii. Calculate bin width and define intervals

B. Equal-Frequency Binning

i. Divide data into **3 bins with equal records**
ii. Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High
- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis *(L3, L4, L5)*

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.
To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

📊 Sample Dataset: Visit Duration

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

✅ Tasks

A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records
ii. Assign each value to its respective bin
iii. Analyze how effectively this method groups short and long visit durations

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B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration
ii. Apply Z-score normalization:

$$Z = \frac{X-\mu}{\sigma}$$

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X-X_{min}}{X_{max}-X_{min}}$$

ii. Transform all values accordingly
iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

Q1. Data Preprocessing in Retail Analytics

A. Handling Missing Values

Missing values reduce data quality and can affect prediction accuracy.

Missing values identified:
• Age: Customer C2
• Monthly_Spend: Customer C3
• Purchase_Frequency: Customer C5

Method 1 – Median Imputation:
Age Median = 36
Monthly_Spend Median = 38
Purchase_Frequency Median = 10

Updated values:
C2 Age = 36
C3 Monthly_Spend = 38
C5 Purchase_Frequency = 10

Method 2 – k-NN Imputation:
This method estimates missing values using similar customer records based on nearest neighbours. It generally provides more realistic values than simple replacement.

Justification:
Median is preferred because the dataset contains outliers (Age=105 and Spend=500). k-NN preserves relationships among attributes.

B. Outlier Detection and Treatment

Outliers are detected using the Z-score method:
Z = (X-Mean)/Standard Deviation

Detected outliers:
• Age = 105
• Monthly_Spend = ₹500

These values belong to Customer C4 and are significantly larger than other observations.

Treatment:
1. Capping (Winsorization): Replace Age 105 with 60 and Spend 500 with 300.
2. Transformation: Apply logarithmic transformation to reduce skewness.
3. Removal: Remove only if confirmed as an incorrect entry.

Capping is recommended because valuable customer information is retained.

C. Categorical Data Standardization

The Gender column contains inconsistent values:
M, Male, male, Female, FEMALE.

Standardized format:
M → Male
Male → Male
male → Male
Female → Female
FEMALE → Female

Standardization improves consistency and avoids duplicate categories.

Q2. Covariance and Data Reduction

A. Covariance Analysis

Covariance measures how two variables change together.

Income and Purchase_Value show a positive covariance because customers with higher income generally make higher purchases.

Interpretation:
• Positive covariance → Both variables increase together.
• Negative covariance → One increases while the other decreases.

Importance:
• Identifies relationships between variables.
• Helps feature selection.
• Improves predictive modelling.

B. Feature Selection

Redundant features:
Income, Purchase_Value, Average_Balance and EMI are strongly related.

Selected features:
Age, Income, Credit_Score, Orders and Customer_Class.

Advantages:
• Reduces computation time.
• Removes redundant information.
• Improves model accuracy.
• Prevents overfitting.

C. Data Transformation

Standardize all numerical attributes except Cust_ID and Customer_Class.

Formula:
$Z=(X-\mu)/\sigma$

Need for standardization:
• Places all features on the same scale.
• Improves distance-based algorithms.
• Speeds up model training.
• Prevents large-valued attributes from dominating.

Q3. Data Transformation and Discretization

A. Equal-Width Binning

Study_Hours:
Minimum=2, Maximum=11
Bin width=(11-2)/3=3

Bins:
Bin1: 2–5
Bin2: 5–8
Bin3: 8–11

Test_Score:
Minimum=40, Maximum=100
Bin width=(100-40)/3=20

Bins:
40–60
60–80

80–100

Equal-width binning divides data into intervals of equal size.

B. Equal-Frequency Binning

Total records=10.

Study_Hours:
Bin1: 2,3,4
Bin2: 5,6,7
Bin3: 8,9,10,11

Test_Score:
Bin1: 40,50,55
Bin2: 65,75,85
Bin3: 90,95,98,100

Each bin contains nearly equal numbers of observations.

C. Concept Hierarchy

Study_Hours:
2–4 → Low
5–7 → Medium
8–11 → High

Test_Score:
40–59 → Poor
60–79 → Average
80–100 → Excellent

Concept hierarchies simplify detailed numerical values into meaningful categories.

Q4. Normalization

Given Data: 200, 300, 400, 600, 1000

Min-Max Normalization:
200→0.00
300→0.125
400→0.25
600→0.50

1000 → 1.00

Mean=500
Standard Deviation≈282.84

Z-score values:
200 → -1.06
300 → -0.71
400 → -0.35
600 → 0.35
1000 → 1.77

Conclusion:
Min-Max scales values between 0 and 1.
Z-score centers data around zero and is more suitable when outliers are present.

Q5. Data Transformation in Online Shopping Behaviour Analysis

Scenario:
The company analyses customer visit duration to improve customer segmentation using clustering algorithms.

Equal-Frequency Binning:
Bin1: 15,30,45
Bin2: 60,120,300
Bin3: 600,900,1800,3600

Interpretation:
Bin1 represents short visits.
Bin2 represents medium visits.
Bin3 represents long visits.

Z-score Normalization:
Mean≈747 seconds
Standard Deviation≈1096 seconds

Example:
15 seconds≈-0.67
3600 seconds≈2.60

Large values receive high positive Z-scores and can be identified as extreme observations.

Min-Max Normalization:
Formula:
(X-Xmin)/(Xmax-Xmin)

15 → 0
120 → 0.029
3600 → 1

Comparison:
Min-Max is simple and produces values between 0 and 1 but is sensitive to outliers.
Z-score is preferred for machine learning because it handles extreme values more effectively.